

$$C_{\phi\text{WB}}^{(22)} \rightarrow$$

$$\begin{aligned} & \frac{1}{16\pi^2} \left(-\frac{1}{8} g_1 g_2 \left(-2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + \sum_{\mathbf{p}} g_2^2 \right) \text{LF}_{3,0}[\mathbf{m}_{\tilde{l}}^{\text{p}}] + \frac{1}{8} g_1 g_2 \left(-2 \overline{y_e^{\text{pr}}} y_e^{\text{pr}} + \sum_{\mathbf{p}} g_2^2 \right) \text{LF}_{4,-1}[\mathbf{m}_{\tilde{l}}^{\text{p}}] + \right. \\ & \frac{1}{8} g_1 g_2 \left(-2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} + \sum_{\mathbf{p}} g_2^2 \right) \text{LF}_{3,0}[\mathbf{m}_{\tilde{q}}^{\text{p}}] - \frac{1}{8} g_1 g_2 \left(-2 \overline{y_d^{\text{pr}}} y_d^{\text{pr}} + \sum_{\mathbf{p}} g_2^2 \right) \text{LF}_{4,-1}[\mathbf{m}_{\tilde{q}}^{\text{p}}] + \\ & \frac{1}{4} g_1 g_2^3 \text{LF}_{2,2,-1}[\mathbf{m}_2, \tilde{\mu}] + \frac{1}{2} g_1 g_2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_{\tilde{l}}^{\text{p}}, \mathbf{m}_{\tilde{e}}^{\text{r}}] - g_1 g_2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_{\tilde{l}}^{\text{p}}, \mathbf{m}_{\tilde{e}}^{\text{r}}] + \\ & \frac{1}{2} g_1 g_2 \overline{a_e^{\text{pr}}} a_e^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_{\tilde{l}}^{\text{p}}, \mathbf{m}_{\tilde{e}}^{\text{r}}] + \frac{1}{2} g_1 g_2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_{\tilde{q}}^{\text{p}}, \mathbf{m}_{\tilde{d}}^{\text{r}}] - \\ & 2 g_1 g_2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_{\tilde{q}}^{\text{p}}, \mathbf{m}_{\tilde{d}}^{\text{r}}] + \frac{3}{2} g_1 g_2 \overline{a_d^{\text{pr}}} a_d^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_{\tilde{q}}^{\text{p}}, \mathbf{m}_{\tilde{d}}^{\text{r}}] + \\ & \frac{1}{4} g_1 g_2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{2,2,0}[\mathbf{m}_{\tilde{q}}^{\text{p}}, \mathbf{m}_{\tilde{u}}^{\text{r}}] + \frac{1}{4} g_1 g_2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_{\tilde{q}}^{\text{p}}, \mathbf{m}_{\tilde{u}}^{\text{r}}] - \\ & \frac{1}{4} g_1 g_2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_{\tilde{q}}^{\text{p}}, \mathbf{m}_{\tilde{u}}^{\text{r}}] - \frac{1}{4} g_1 g_2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_{\tilde{q}}^{\text{p}}, \mathbf{m}_{\tilde{u}}^{\text{r}}] + \\ & \frac{3}{4} g_1 g_2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,1,0}[\mathbf{m}_{\tilde{u}}^{\text{r}}, \mathbf{m}_{\tilde{q}}^{\text{p}}] - \frac{1}{4} g_1 g_2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{3,2,-1}[\mathbf{m}_{\tilde{u}}^{\text{r}}, \mathbf{m}_{\tilde{q}}^{\text{p}}] - \\ & \frac{9}{4} g_1 g_2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{4,1,-1}[\mathbf{m}_{\tilde{u}}^{\text{r}}, \mathbf{m}_{\tilde{q}}^{\text{p}}] + \frac{3}{2} g_1 g_2 \tilde{\mu}^2 \overline{y_u^{\text{pr}}} y_u^{\text{pr}} \text{LF}_{5,1,-2}[\mathbf{m}_{\tilde{u}}^{\text{r}}, \mathbf{m}_{\tilde{q}}^{\text{p}}] - \\ & \frac{1}{4} g_2 g_1^3 \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_1] + \frac{3}{4} g_2 g_1^3 \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_1] - \frac{1}{2} g_2 g_1^3 \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_1] - \\ & \left. \frac{7}{4} g_1 g_2^3 \text{LF}_{3,1,-1}[\tilde{\mu}, \mathbf{m}_2] + \frac{13}{4} g_1 g_2^3 \text{LF}_{4,1,-2}[\tilde{\mu}, \mathbf{m}_2] - \frac{3}{2} g_1 g_2^3 \text{LF}_{5,1,-3}[\tilde{\mu}, \mathbf{m}_2] \right) \end{aligned}$$